

Vithia 0-Vita-1: A Deterministic Context Protocol for Verifiable Scientific State Transitions

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Abstract

Scientific workflows can conflate evidence with interpretation, prediction with observation, provenance with correctness, and graph connectivity with causality. We present 0-Vita-1, a Vithia protocol separating deterministic context compilation from downstream decisions while preserving typed provenance and append-only custody. Simulated experiments test exact addressability (E0), rule inference under corruption (E1), history-sensitive context (E2), uncertainty over hidden deterministic worlds (E3), and a frozen Space Invaders snapshot set (E4A). E0, E2, and E3 are supported; E1 is partial and retains negative/null outcomes; E4A contains no decider calls. Breakpoint roots and an ordered MMR preserve identity and order, not truth. Each committed state is finite while successor context is indefinitely extensible.

1 Introduction

0-Vita-1 treats scientific type errors as a protocol problem: exact evidence identity remains separate from contextual interpretation, and Vithia-S0 context compilation remains separate from System-1 decision-making. Hashes identify exact bytes; typed FCG edges declare relationships; breakpoint roots commit governed artifact sets; and an ordered MMR commits their sequence. These integrity mechanisms do not establish scientific truth.

2 Formal architecture

$$E_t \xrightarrow{S_0} C_t \xrightarrow{\Pi} P_t \xrightarrow{S_1} D_t \xrightarrow{\text{world}} O_{t+1} \xrightarrow{S_0} C_{t+1}.$$

Vithia-S0 compiles source evidence into finite VitaState context. A bounded ContextPacket is projected for one downstream question; a subsequent observation creates successor evidence.

0-Vita-1 evidence/context/decision loop



Figure 1: 0-Vita-1 evidence, context, decision, and successor loop

identity $\neq$ meaning, custody $\neq$ correctness, prediction $\neq$ observation.

3 Mechanical Scientific Method

The execution chain is question, hypothesis, preregistration, input freeze, execution, observation, analysis, claim decision, breakpoint, independent verification, and successor. Historical failures, nulls, and abstentions remain addressable.

Identity, relationships, and append-only custody

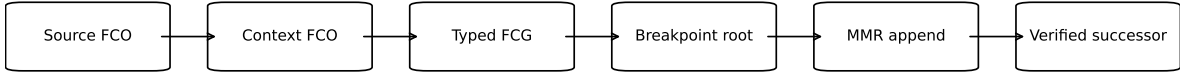


Figure 2: FCO, typed FCG, breakpoint, and ordered MMR custody

4 Experiment ladder

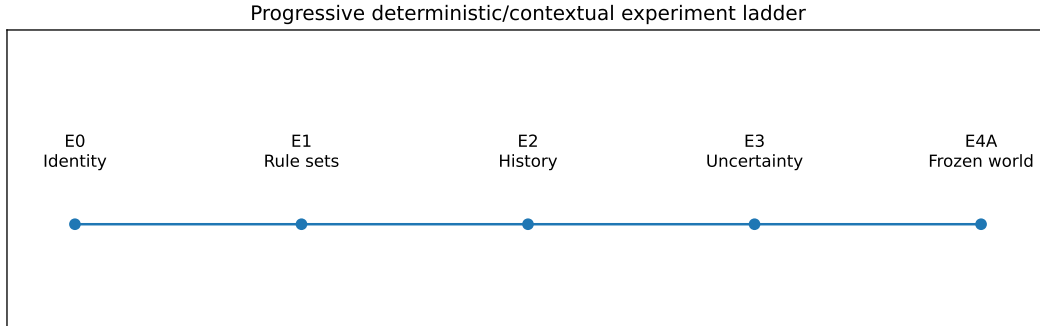


Figure 3: Progressive experimental burden from identity to partial observation

E1 uses CellPyLib as an independent cellular-automaton implementation comparison [2]; E3 uses a separately implemented SAT-oracle lane based on PySAT [3]; E4A uses the Arcade Learning Environment as the single-player partially observed game substrate [1].

5 Results

5.1 E0: exact addressability

E0 is SUPPORTED. Across 10,010 rows the receipt records zero round-trip failures, zero address collisions, zero content-ID collisions, and zero address/hash-equality rows. Replay is Level 4.

5.2 E1: rule inference

The independent comparison agreed on 100,352 cells with zero mismatches. E1 remains PARTIAL: H1d rank-1 fraction 0.806 is below the 0.90 threshold; Wilson 95% CI [0.777,0.832]. H1e ($p=1$) and H1e-v2 ($p=0.559$) both FAIL_TO_REJECT_H0.

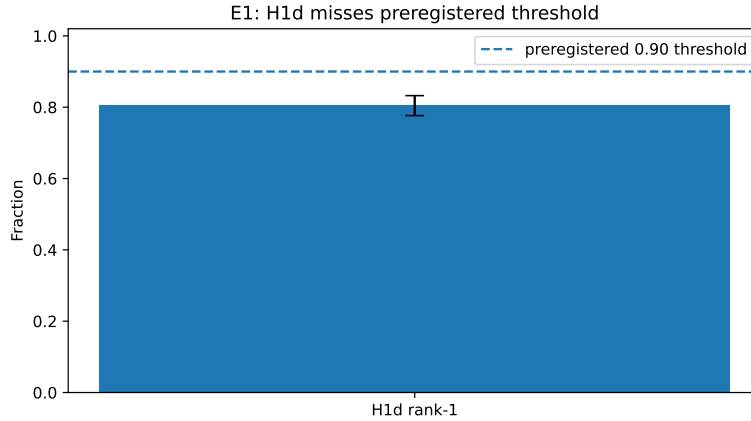


Figure 4: E1 H1d and the preregistered threshold

5.3 E2: history-sensitive context

E2 is SUPPORTED. Correct-history accuracy 1.000, shuffled-history 0.915, and no-history unknown rate 1.000. There are 57 same-current-atom/different-context cases. Secondary matched accuracy delta 0.087, 95% bootstrap CI [0.019,0.163].

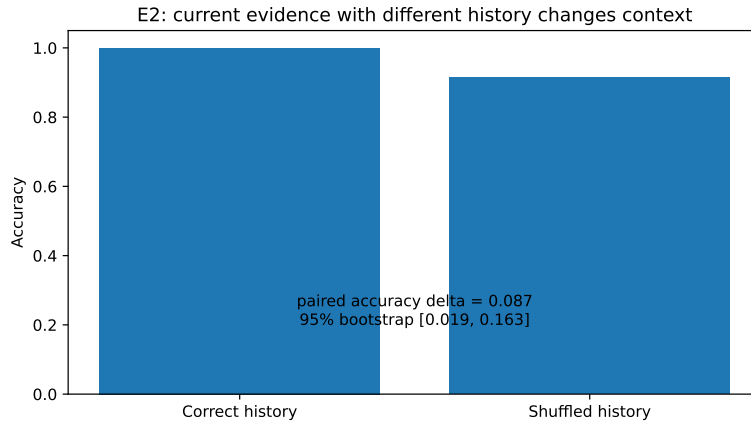


Figure 5: E2 history manipulation

5.4 E3: hidden deterministic worlds

E3 is SUPPORTED with abstention. Kernel and SAT oracle agree on 28,185 eligible cells with zero disagreements; 66 states (6,484 cells) are ABSTAIN_SIZE_LIMIT. For 200 pairs, max-EIG realized gain is 1.9227 bits versus 1.7175 random ($p=0.00605$); secondary paired delta 0.205 bits, 95% CI [0.038,0.377]. Replay remains Level 2.

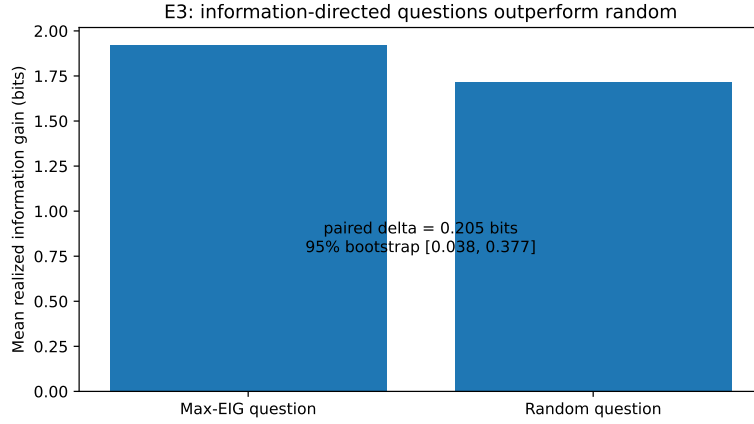


Figure 6: E3 information-directed versus random questions

5.5 E4A: frozen partial observation

E4A freezes 96 replay-verified snapshots balanced 48/48 between bomb-positive and bomb-negative strata. It is INPUT_FROZEN_NO_DECIDER_CALLS. No JEV, OpenJEV, Liquid, or Ollama/Ollarma utility claim follows.

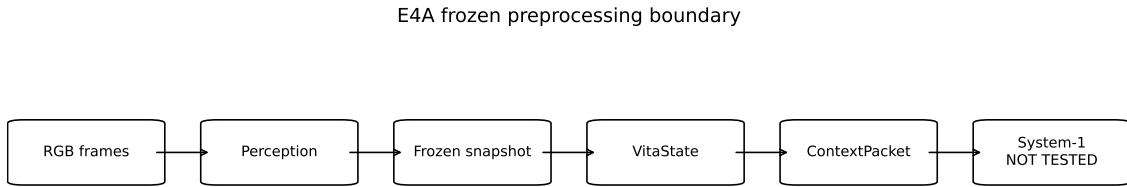


Figure 7: E4A boundary; downstream System-1 is not tested

6 Post-confirmatory statistics

Secondary analyses use frozen outputs and do not alter confirmatory states. E3 paired Cohen d-z is 0.168; E3 abstention fraction is 0.180.

7 Reproducibility and custody

The publication evidence lock references UFA-JEV-BP-0024, scientific MMR size 24, root a461def1576e...d76692. The complete root is retained in the governed evidence lock and supplement. Manuscript evolution uses a distinct publication MMR.

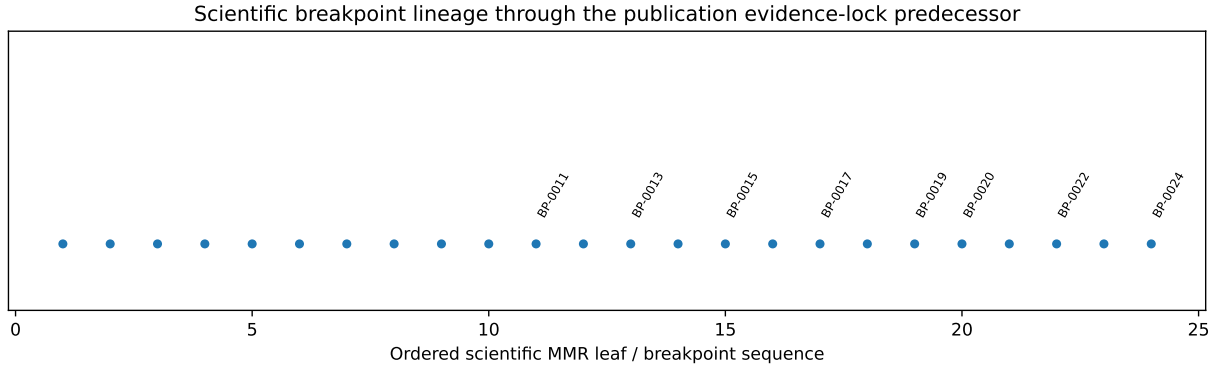


Figure 8: Ordered scientific breakpoint lineage

8 Discussion

The results support a narrow architectural conclusion: exact evidence can remain immutable while deterministic rules produce versioned context and uncertainty remains explicit. Deterministic evidence handling does not imply a deterministic downstream model or world. Artificial Infinite Systems is structural rather than literal: every committed state is finite, while successor states can extend without rewriting predecessors.

9 Limitations

DeltaGStar is NOT_COMPUTED; private DeltaGStar and Anticube mathematics are not disclosed and no surrogate is labeled DeltaGStar. Cross-host replication is DEFERRED_NOT_FAILED because magicPRObox was unavailable. E3 is Level-2 replay. Hosted JEV, OpenJEV, System One, Liquid, and Olama/Ollama utility are NOT_TESTED. All evidence is simulated; biological transfer is NOT_TESTED. Signatures are NOT_SIGNED.

10 Conclusion

0-Vita-1 provides a testable boundary between immutable evidence, deterministic context compilation, explicit uncertainty, downstream decisions, and successor observations. v0.1.0 establishes the substrate through E4A while preserving negative, null, partial, abstention, and not-tested states.

References

- [1] Bellemare et al. The Arcade Learning Environment. JAIR 47 (2013). doi:10.1613/JAIR.3912.
- [2] Antunes. CellPyLib. JOSS 6(67) (2021). doi:10.21105/joss.03608.
- [3] Ignatiev et al. PySAT. SAT 2018. doi:10.1007/978-3-319-94144-8_26.